

CSC1003 Practice 4

Conditionals, Loops and Arrays

Overview

- Data Structure
 - Why is data structure important?
- Array
 - Creating an array
 - Accessing data in an array
- Practice

40	55	63	17	22	68	89	97	89
0	1	2	3	4	5	6	7	8

<- Array Indices

Array Length = 9
First Index = 0
Last Index = 8

Why are data structures useful?

Consider your first assignment. The first problem is about the Fibonacci series. If you don't use data structures like arrays, it's hard to print the elements from the end. Data structures are useful in real-life situations.

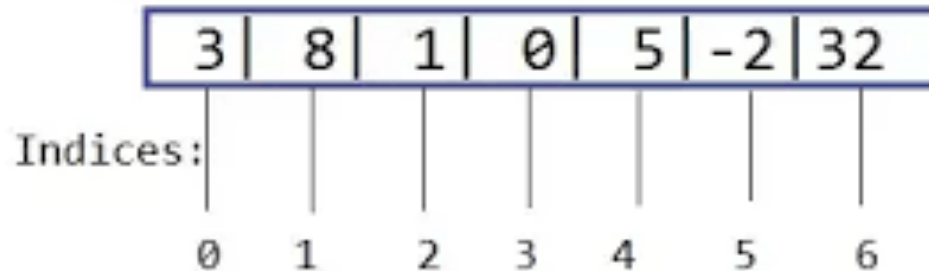
```
int[] fibonacci_series = new int[n];
for (int i=0; i<n; i++) {
    fibonacci_series[i] = fibonacci_series[i-1]+fibonacci_series[i-2];
}
for (int i=n-1; i>n-d; i--){
    System.out.println(fibonacci_series[i]);
}
```

Arrays

A new variable type is an object that represents an ordered, homogeneous list of data.

- Arrays have many elements that you can access using indices

Array :



Creating an array

```
type[] name = new type[length];
```

```
int[] numbers = new int[5]; //Create an array with five integer  
elements
```

index	0	1	2	3	4
value	0	0	0	0	0

Accessing Data In An Array

```
name[index] // get element at index
```

- Indices go from 0 to the array's length - 1.

```
for (int i = 0; i < 7; i++) {  
    println(numbers[i]);  
}  
println(numbers[9]); // exception  
println(numbers[-1]); // exception
```

<i>index</i>	0	1	2	3	4	5	6
<i>value</i>	0	1	2	3	4	5	6

Putting data in an array

```
name[index] = value; // set element at index
```

- Like Strings, indices go from 0 to the array's length - 1.

```
int[] numbers = new int[7];  
for (int i = 0; i < 7; i++) {  
    numbers[i] = i;  
}
```

Index	0	1	2	3	4	5	6
Value	0	1	2	3	4	5	6

Array of other types

```
double[] results = new double[5];
```

```
String[] names = new String[3];
```

```
boolean[] switches = new boolean[4];
```

- Java initializes each element of a new array to its default value, which is **0** for int and doubles, **'\0'** for char, **false** for boolean, and **null** for objects.

Array length

You can get the length of an array by saying
`myArray.length`

Practice:

What is the index of the last element of an array in terms of its length?

What is the index of the middle element of an array in terms of its length

Practice: what is the output of the following program and why?

```
1  public class Test {  
    Run | Debug  
2      public static void main(String[] args) {  
3          int[] a = {1, 2, 3, 4};  
4          int[] b = new int[4];  
5          int[] c = new int[5];  
6          b = a;  
7          c = a;  
8          b[1] = 0;  
9          c[2] = 0;  
10         c = new int[3];  
11         for (int i : a) {  
12             System.out.println(i);  
13         }  
14         for (int i : c) {  
15             System.out.println(i);  
16         }  
17     }  
18 }
```

Solution

1. Initialization:

- `a` is initialized as `{1, 2, 3, 4}`.

2. Assignment:

- `b = a;` and `c = a;` make both `b` and `c` reference the same array as `a`.
- So, `a`, `b`, and `c` all point to `{1, 2, 3, 4}`.

3. Modification:

- `b[1] = 0;` Now, the array is `{1, 0, 3, 4}`.
- `c[2] = 0;` Now, the array is `{1, 0, 0, 4}`.

4. Reassignment:

- `c = new int[3];` reassigns `c` to a new array of size 3, initialized to `{0, 0, 0}`.
- Now, `c` no longer references the same array as `a` and `b`.

5. Printing:

- The first loop prints the elements of `a`: `{1, 0, 0, 4}`.
- The second loop prints the elements of `c`: `{0, 0, 0}`.

Prime Counter

- Write a program PrimeCounter that takes an integer n and finds the number of primes less than or equal to n .
- Note: If you are not careful, your program may not finish in a reasonable amount of time! Try to find a fast way to solve this problem.
- Hint: If a number n can be obtained by multiplying two numbers a and b , i.e., $n = a \times b$, then at least one factor, a or b , must be less than or equal to the square root of n .
- (Optional: To give an efficient program, you might need to use a boolean array.)
- OJ link: https://oj.cuhk.edu.cn/d/csc1003_2025_fall/p/P1009

Q&A